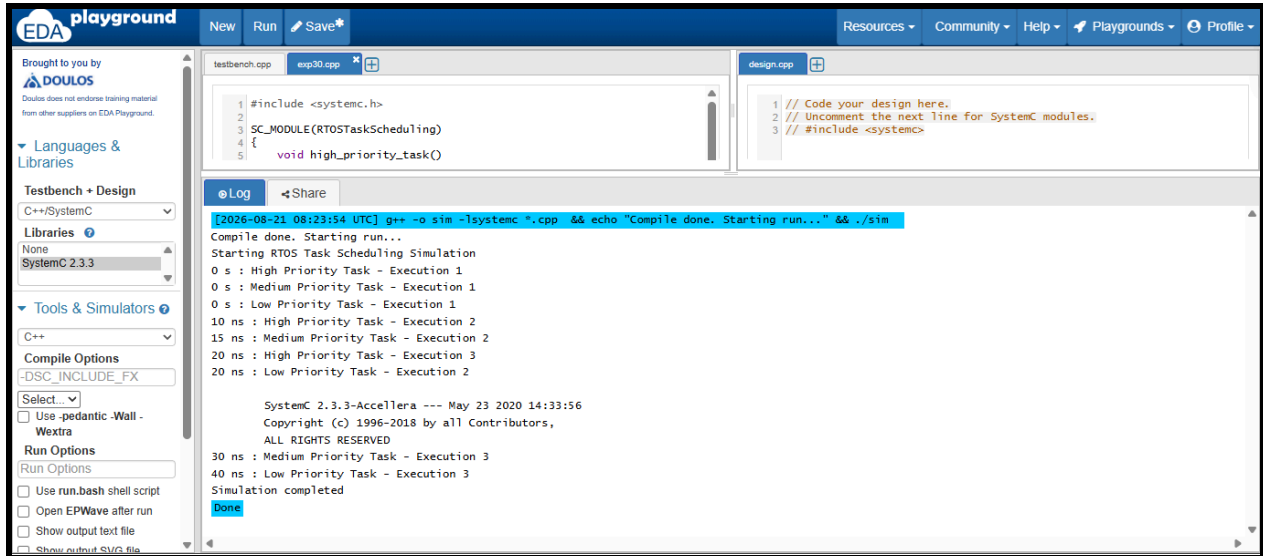




C++ Code:

```
#include <systemc.h>

SC_MODULE(RTOSTaskScheduling)
{
    void high_priority_task()
    {
        for(int i = 1; i <= 3; i++)
        {
            cout << sc_time_stamp()
                << " : High Priority Task - Execution "
                << i << endl;

            wait(10, SC_NS);
        }
    }

    void medium_priority_task()
    {
        for(int i = 1; i <= 3; i++)
        {
            cout << sc_time_stamp()
                << " : Medium Priority Task - Execution "
                << i << endl;

            wait(15, SC_NS);
        }
    }

    void low_priority_task()
    {
        for(int i = 1; i <= 3; i++)
        {
            cout << sc_time_stamp()
                << " : Low Priority Task - Execution "
                << i << endl;

            wait(20, SC_NS);
        }
    }
}
```

```

{
    for(int i = 1; i <= 3; i++)
    {
        cout << sc_time_stamp()
            << " : Low Priority Task - Execution "
            << i << endl;

        wait(20, SC_NS);
    }
}

SC_CTOR(RTOSTaskScheduling)
{
    SC_THREAD(high_priority_task);
    SC_THREAD(medium_priority_task);
    SC_THREAD(low_priority_task);
}
};

int sc_main(int argc, char* argv[])
{
    RTOSTaskScheduling rtos("RTOS");

    cout << "Starting RTOS Task Scheduling Simulation"
        << endl;

    sc_start(70, SC_NS);

    cout << "Simulation completed"
        << endl;

    return 0;
}

```